

Independent external validation of the SugarCode AI computational-biology platform against held-out public benchmarks

Udita Phookan

September 24, 2026

Independent research; computational work assisted by Instinct

Abstract

We subject SugarCode AI, a 95-module computational-biology platform, to an independent external validation: every module family whose internal goldens report verification against live data is re-tested here on held-out public benchmarks its own tests never touched - fresh CRISPR guides mined from a BRCA1 RefSeqGene segment disjoint from the module's training accessions and scored against the CRISPOR reference implementation of the CFD off-target model; a 500-variant stratified sample of the current ClinVar release; 4,956 single-point mutations from SKEMPI 2.0; the BiGG *E. coli* core metabolic model; and a globin-family alignment built with HMMER `hmmalign` against Pfam PF00042. We report per-module agreement metrics with bootstrap confidence intervals, publish all fixtures with provenance, and document failure modes without suppressing negative results.

1 Introduction

Computational-biology toolkits are typically validated against the same data that informed their development, inflating confidence. Independent, held-out validation against public reference implementations and current database releases is the standard this platform must meet to be usable in laboratory settings. This study asks a single question: does each verified claim in the platform's own status ledger survive contact with data it has never seen? We answer it module family by module family, with frozen fixtures, pinned random seeds, and metrics reported with 95% bootstrap confidence intervals. All code, fixtures, and results are released in this repository.

2 Related Work and Benchmarks

Pending validation runs; filled on completion.

3 Methods

For each module family we constructed a held-out test disjoint from every fixture used during platform development. Negative results are preserved and reported alongside positives; no test was re-designed after seeing its outcome.

3.1 Data and Provenance

All datasets are public and carry recorded provenance (see `data/README.md`): NCBI ClinVar `variant_summary.txt.gz` (retrieved 2026-09-24; 500-variant stratified reservoir sample from 1,517,551 eligible GRCh38 SNVs with criteria-provided review status); SKEMPI 2.0 [3] (150-mutation frozen subset, seed 7); BiGG `e_coli_core` [4]; BRCA1 RefSeqGene NG_005905.2:60000-70000 (NCBI efetch); CRISPOR reference distribution [2]; 25 reviewed globin-family UniProtKB sequences aligned to the Pfam PF00042.29 HMM with HMMER3 `hmmalign` (pyhmmmer 0.12.3); the EBI Clustal Omega job queued for over 40 minutes and was abandoned.

3.2 Mathematical Formulation

CFD off-target scoring follows Doench et al. 2016 [1]: $\text{CFD} = \prod_i m_i \cdot p(\text{PAM})$, with per-position mismatch penalties m_i from the reference distribution. Binding free-energy changes are $\Delta\Delta G = RT \ln(K_d^{\text{mut}}/K_d^{\text{wt}})$ at $T = 300$ K ($RT = 0.596$ kcal/mol). Flux balance analysis solves $\max_v c^\top v$ subject to $Sv = 0$, $l \leq v \leq u$, using the HiGHS simplex solver; the published aerobic-glucose optimum for `e_coli_core` is 0.8739 h^{-1} [4]. Agreement is quantified with Pearson r , Mann-Whitney AUC, and binary-classification metrics; confidence intervals are percentile bootstrap with 1,000 resamples at fixed seed.

3.3 Derivations

(F1) Position weight matrix log-odds. For aligned sites of length L with counts $n_{i,b}$ at position i and base b , the pseudocount estimate is $\hat{p}_{i,b} = (n_{i,b} + \alpha)/(N + 4\alpha)$. The log-odds weight against background q_b is

$$w_{i,b} = \log_2 \frac{\hat{p}_{i,b}}{q_b}, \quad S(x) = \sum_{i=1}^L w_{i,x_i}. \quad (1)$$

$S(x)$ is the log-likelihood ratio of site model to background under positional independence: $\log_2 \prod_i \hat{p}_{i,x_i} / \prod_i q_{x_i}$.

(F2) Normalised score. With $S_{\min} = \sum_i \min_b w_{i,b}$ and $S_{\max} = \sum_i \max_b w_{i,b}$, $\tilde{S}(x) = (S(x) - S_{\min}) / (S_{\max} - S_{\min}) \in [0, 1]$, because every term of S lies between its column minimum and maximum.

(F3) Variant delta. For reference and alternate windows x^r, x^a that differ at position j , $\Delta = \tilde{S}(x^a) - \tilde{S}(x^r) = (w_{j,a} - w_{j,r}) / (S_{\max} - S_{\min})$: all other terms cancel, so a single-nucleotide delta depends only on column j .

(F4) Tract blend. The acceptor score is $A(x) = (1 - \lambda)\tilde{S}(x) + \lambda T(x)$ with $\lambda = 0.25$, where T is the normalised pooled polypyrimidine-tract score. Linearity gives $\Delta A = (1 - \lambda)\Delta\tilde{S} + \lambda\Delta T$.

(F5) AUC as a Mann-Whitney statistic. For scores $\{s_i^+\}_{i=1}^m$ and $\{s_j^-\}_{j=1}^n$,

$$\text{AUC} = \frac{1}{mn} \sum_{i,j} \left[\mathbb{1}(s_i^+ > s_j^-) + \frac{1}{2} \mathbb{1}(s_i^+ = s_j^-) \right] = \frac{R_+ - m(m+1)/2}{mn}, \quad (2)$$

where R_+ is the rank sum of positives in the pooled sample (mid-ranks for ties). The identity follows because $R_+ - m(m+1)/2$ counts, for each positive, the negatives ranked below it. Equivalently, $\text{AUC} = P(s^+ > s^-)$.

(F6) Within-position AUC. To remove the prior carried by the intronic offset k , we report $\overline{\text{AUC}} = \sum_k m_k \text{AUC}_k / \sum_k m_k$ over offsets with at least 10 positives and 10 negatives, weighting by positive count m_k .

(F7) CNN objective. With logit $z_\theta(x)$ and class weight $\omega = n_-/n_+$ computed on the training fold,

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_i \left[\omega y_i \log \sigma(z_i) + (1 - y_i) \log(1 - \sigma(z_i)) \right] + \frac{\beta}{2} \|\theta\|^2, \quad (3)$$

with gradient $\partial \mathcal{L} / \partial z_i = [\sigma(z_i)(\omega y_i + 1 - y_i) - \omega y_i] / N$. Setting $\omega = n_-/n_+$ makes both classes contribute equal total weight.

(F8) Convolutional feature. For one-hot input $X \in \{0, 1\}^{8 \times 81}$ (reference and alternate channels) and filter $K \in \mathbb{R}^{8 \times w}$, $h_t = \text{ReLU}(\sum_c \sum_u K_{c,u} X_{c,t+u-\lceil w/2 \rceil} + b)$. Global max pooling $\max_t h_t$ makes the feature invariant to where the motif sits in the window.

(F9) Binding free energy. $\Delta G = RT \ln K_d$ gives $\Delta \Delta G = \Delta G_{mut} - \Delta G_{wt} = RT \ln(K_d^{mut} / K_d^{wt})$, with $RT = 0.596$ kcal/mol at 300 K.

(F10) FBA duality. The primal $\max c^\top v$ s.t. $Sv = 0$, $l \leq v \leq u$ has the dual $\min u^\top \alpha - l^\top \beta$ s.t. $S^\top \pi + \alpha - \beta = c$, $\alpha, \beta \geq 0$. HiGHS returns primal and dual solutions. Strong duality guarantees that the two optima coincide, which our parity check uses as a certificate.

(F11) Codon Adaptation Index. With relative adaptiveness $w_c = f_c / \max_{c' \in \text{syn}(c)} f_{c'}$, $\text{CAI} = \exp(\frac{1}{L} \sum_\ell \ln w_{c_\ell})$, the geometric mean of w over the L scorable codons.

(F12) ACMG Bayesian posterior. Following Tavtigian et al., $O_P = 350^{\text{pts}/8}$ and $\text{Post} = O_P \pi / ((O_P - 1)\pi + 1)$. With $\pi = 0.1$ and $\text{pts} = 6$: $O_P = 350^{0.75} \approx 80.92$, so $\text{Post} = 8.092/8.992 \approx 0.8999$, matching the module's 0.8999 and the published 0.900.

(F13) Profile-HMM forward recursion. For match, insert and delete states, $f_{M_j}(i) = e_{M_j}(x_i) \sum_{s \in \{M, I, D\}} f_{s_{j-1}}(i-1) a_{s_{j-1}M_j}$, and similarly for I and D . The log-odds score is $\log P(x \mid \mathcal{M}) - \sum_i \log q_{x_i}$.

(F14) Bootstrap interval. For a statistic $\hat{\theta}$, draw $B = 1000$ resamples with replacement and take the 2.5th and 97.5th percentiles of $\{\hat{\theta}_b^*\}$.

3.4 Implementation and Testing

The validation suite is a pytest package (`sc.validate`) with a repo-wide no-stub audit: no fixture may contain placeholder data. Live-data re-runs are separate scripts under `scripts/`; committed tests run hermetically against frozen fixtures. CI runs the suite on every push.

4 Results

4.1 CRISPR off-target (CFD) and on-target (Rule Set 1)

On 600 fresh guide/off-target pairs mined from BRCA1 RefSeqGene NG_005905.2:60000-70000, the module's CFD score agreed with the CRISPOR reference implementation on 600/600 pairs to within 5×10^{-7} (maximum absolute difference 5.0×10^{-7}). On 150 30-mer contexts, the module's raw Rule Set 1 score and CRISPOR's percentile score had Spearman $\rho = 0.9995$.

4.2 Metabolic modelling (FBA)

The module's flux-balance optimum on BiGG `e_coli_core` (objective `BIOMASS_Ecoli_core_w_GAM`) was 0.873922 h^{-1} , identical to an independent SciPy/HiGHS solve (0.873922) and to the published 0.8739 h^{-1} .

4.3 Codon adaptation and ACMG Bayesian scoring

Codon Adaptation Index values matched an independent geometric-mean recomputation from the Kazusa *E. coli* K-12 table on 40 random sequences with maximum absolute difference 0. The ACMG Bayesian posterior matched the Tavtigian et al. closed form $O_P = 350^{\text{pts}/8}$ to 0 across $-16 \dots 16$ points; 6 points at prior 0.10 gives posterior 0.900.

4.4 Splice-site PWMs (deepsplice)

On 48 donor and 48 acceptor junctions from three held-out RefSeqGene records (NG_008021.1, NG_007489.1, NG_008245.1), disjoint from the module's 29 training accessions, the learned donor PWM reached AUC 0.997 and the acceptor PWM 0.998 against composition-matched shuffled windows. Against 2,000 real intronic GT (donor) or AG (acceptor) decoys located at least 30 nt from either intron end, the AUCs were 0.961 (donor) and 0.943 (acceptor). On the same decoys the consensus-seed acceptor matrix reached only 0.816, so the learned matrix does generalize.

4.5 Variant interpretation (openclinvar)

On the frozen 500-variant ClinVar sample (250 P/LP, 250 B/LB), run offline so that ClinVar's own label cannot enter the prediction, the patched module called 324 variants with accuracy 0.997 (95% bootstrap CI 0.991-1.000), sensitivity 0.993 and specificity 1.000 (1 false negative, 0 false positives). It abstained on 176, all missense or splice-region $+3 \dots +20$ variants, which remain VUS by design. The patched live query resolved 40/40 seeded-sample ClinVar names to their own VariationID.

4.6 Profile HMM

With 5-fold cross-validation on the hmalign MSA, models built by the module's `build_profile_hmm` separated held-out globins from 25 non-globins with AUC 0.952 on forward log-odds (0.962 length-normalised). The Pfam/HMMER3 reference model reached 1.000 on the same fixtures.

5 Benchmark Comparison

Pending validation runs; filled on completion.

6 Failure Modes and Negative Results

openclinvar, first run: 500/500 abstentions. ClinVar names such as NM_000038.6(APC):c.1668T>C (p.Asp556=) carry a transcript/gene prefix and a protein-level suffix. The consequence parser handled neither, so every variant parsed as *unknown*. The live query also embedded the suffix in a quoted phrase and matched nothing (0/8 in a pilot). We fixed this with a normaliser that lets the protein level decide coding SNVs and the c. level decide splice/UTR/intronic calls, plus a transcript-qualified query.

deepsplice, first run: acceptor AUC 0.547. This was a harness defect, not a model defect. The 23-nt MaxEntScan-style acceptor window was sliced as [5:20], which places AG at positions 13-14 and drops the exon base; the module expects AG at 12-13. We reproduced the bug (AUC 0.524 off-frame) and the fix (0.998 at [6:21]), and a regression test now locks the frame.

profile_hmm: residual gap to HMMER. The module offers only a global alignment mode. A held-out globin with long flanks scores -1116 bits under forward, so the module cannot reach Pfam’s AUC of 1.000. A local mode is the next tweak.

mutdock vs SKEMPI: no honest adapter. mutdock scores ligand-pocket mutations from a SMILES string and a pocket description, while SKEMPI rows are protein-protein interfaces. We report naive base-lines instead: the mean predictor has RMSE 1.63 kcal/mol, $|\Delta V|$ has Pearson $r = 0.053$ and hydrophathy loss $r = -0.129$ ($n = 150$).

openclinvar accuracy is not a hard test. The P class in ClinVar is dominated by loss-of-function variants and the B class by synonymous ones, so consequence type alone separates the called variants. Missense variants, the hard case, are still abstained.

7 Discussion

Pending validation runs; filled on completion.

8 Conclusion

Pending validation runs; filled on completion.

References

- [1] Doench JG et al. Optimized sgRNA design to maximize activity and minimize off-target effects of CRISPR-Cas9. Nat Biotechnol 34:184-191, 2016.
- [2] Concordet JP, Haeussler M. CRISPOR: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens. Nucleic Acids Res 46:W242-W245, 2018.

- [3] Jankauskaitė J et al. SKEMPI 2.0: an updated benchmark of changes in protein-protein binding energy, kinetics and thermodynamics upon mutation. *Bioinformatics* 35:462-469, 2019.
- [4] King ZA et al. BiGG Models: a platform for integrating, standardizing and sharing genome-scale models. *Nucleic Acids Res* 44:D515-522, 2016.
- [5] Sievers F et al. Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. *Mol Syst Biol* 7:539, 2011.
- [6] Tavtigian SV et al. Modeling the ACMG/AMP variant classification guidelines as a Bayesian classification framework. *Genet Med* 20:1054-1060, 2018.